PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: HalalChain Agent

Version: 2.0 (Hackathon MVP)

Target Domain: Domain 1 - AI Systems & Agentic Workflow Automation

1. Project Overview

- **Problem Statement:** Securing JAKIM (Halal) certification is a highly manual, fragmented, and unstructured workflow for Malaysian SMEs. Non-compliance officers often struggle to cross-reference messy documents, normalize local ingredient names, identify ambiguous sources, and coordinate back-and-forth communication with suppliers. This results in high inefficiency, human error, and frequent application rejections by JAKIM due to missing or improperly formatted data.
- **Target Domain:** Enabling rapid, automated regulatory compliance and stateful workflow orchestration for early-stage SMEs and food producers.
- **Proposed Solution Summary:** HalalChain Agent removes the friction of compliance by enabling an SME to upload messy, unstructured data and receive a structured, ready-to-submit JAKIM application. Using Gemini (via Vercel AI SDK) as a central reasoning engine, the system acts as a stateful workflow orchestrator. It automates document classification, normalizes and verifies ingredient halal status, handles edge cases via supplier verification loops, and provides context-aware chat assistance. This shifts the SME's role from a manual "data compiler" to an "application reviewer."

2. Background & Business Objective

- **Background of the Problem:** For decades, Halal certification preparation has followed a manual, linear pathway. A food producer gathers physical or digital files, manually drafts an ingredient list, researches each item's halal status, and emails suppliers for certificates

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if an ingredient's source is unknown. If they use Manglish or local terms (e.g., writing "minyak sapi" instead of standardizing to "beef tallow"), the application is often delayed or rejected.

- **Business Objective:** To drastically reduce the time, cost, and friction required for SMEs to prepare JAKIM certification applications. By automating gap analysis and supplier coordination, the system ensures 100% document completeness and ingredient transparency prior to final submission, minimizing JAKIM rejection rates.

3. Product Purpose

The purpose of the HalalChain Agent is to act as an intelligent, adaptive workflow copilot. It does not just process data in a vacuum; it maintains the context of the user's entire application state, identifies missing puzzle pieces, and orchestrates the necessary actions (either from the SME, the AI, or a third-party Supplier) to complete the certification puzzle.

3.1. Main Goal of the System: The main goal of the HalalChain Agent is to **automate and streamline the highly manual JAKIM Halal certification preparation process using an AI-agentic workflow.** Rather than SMEs manually cross-referencing messy documents and guessing ingredient compliance, the system acts as an intelligent workflow orchestrator. Its objective is to transform unstructured, messy user data into a 100% compliant, structured, and ready-to-submit Halal application. By automatically identifying compliance gaps, normalizing local ingredients, and orchestrating supplier verification, the system eliminates human error, drastically reduces preparation time, and minimizes application rejections by JAKIM. Ultimately, it shifts the user's role from a frustrated "data compiler" to a confident "application reviewer."

3.2. Intended Users (Target Audience):

- SME Food Producers & Founders
- Ingredient Suppliers

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4. System Functionalities

4.1. Description: The system operates as a stateful, AI-driven workflow orchestrator for Halal compliance. Using Gemini as the central reasoning engine, the system ingests unstructured data (messy files, Manglish ingredient lists), structures it via strict JSON schemas, cross-references for compliance gaps, and orchestrates dynamic resolution loops (manual overrides and supplier checks) to produce a 100% compliant, ready-to-submit certification bundle.

4.2. Key Functionalities:

- **Smart Document Auto-Sort:** Translates unstructured, poorly named file uploads into a categorized, high-fidelity compliance checklist based on JAKIM requirements.
- **Ingredient Verification & Normalization:** Parses unstructured text, normalizes Manglish/local terms to scientific standards, and evaluates Halal status with a confidence score.
- **Agentic Resolution Workflows:** Automatically identifies "Ambiguous" items or missing files and orchestrates a multi-step escalation path to either the SME or a third-party Supplier Portal for resolution.
- **Context-Aware Chatbox:** Facilitates continuous guidance by reading the current application state and suggesting context-specific Halal-certified alternatives for rejected ingredients.
- **Structured Export:** Compiles the verified state data into machine-readable JSON and human-readable PDF audit reports.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Google Gemini (gemini-1.5-flash). *Justification:* With the hackathon constraint dropped, Gemini 1.5 Flash was selected for its massive context window (critical for ingesting multiple user documents), extreme speed, and native compatibility with the Vercel AI SDK's generateObject tool, which guarantees the strict JSON outputs required to power our stateful UI components.

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4.3.2. **Prompting Strategy:** Multi-step agentic prompting combined with strict Zod output schemas. *Reasoning:* A zero-shot approach cannot handle complex regulatory edge cases. The AI must first parse, then translate (Manglish to English), and finally evaluate Halal status. We use multi-step system prompts to force the AI through this chain of thought before outputting the final JSON state.

4.3.3. **Context & Input Handling**: The system extracts raw text from uploaded files locally via the browser's FileReader to prevent oversized payloads. While Gemini handles 1M+ tokens, we implement client-side chunking/truncation if a document exceeds 50 pages. If an input is entirely unreadable, the system triggers a graceful rejection prompting the user to manually input their ingredient list.

4.3.4. **Fallback & Failure Behavior:** To combat hallucination, we implemented strict **Confidence Thresholds**. If the model returns a decision with <60% confidence, the system deliberately halts the automated workflow, triggers a "Manual Review" warning state, and forces human escalation. If the AI fails to generate the correct JSON structure, the Vercel AI SDK automatically retries the generation.

5. User Stories & Use Cases

- **User Stories:** *"As an SME founder, I want to drag and drop my messy files into the system so that the AI automatically sorts them and tells me exactly which mandatory documents I am missing. " "As an Ingredient Supplier, I want to log into a portal to see exactly which ingredients my clients need verified, so I can upload my Halal Certificate once and unblock their application.""*
- **Use Cases (Main Interactions):**

  Document Upload & Gap Analysis Flow: User uploads files. The AI parses the text, classifies them (e.g., premises-license), and updates the UI checklist. If gaps exist, the "Export" button remains locked.

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Ingredient Resolution Flow: User inputs ingredients. The AI normalizes the terms and flags an ingredient as "Ambiguous" (e.g., Emulsifier). The user clicks "Request Supplier Verification," triggering the Supplier loop.

Supplier Escalation Flow: Supplier logs into their view, sees the pending request, marks it as "Halal," and uploads a mock certificate. This state instantly syncs back to unblock the SME.

6. Features Included (Scope Definition)

A drag-and-drop document classification engine and visual compliance gap checker.

Automated Manglish-to-English ingredient normalization and Halal status scoring via Gemini.

A mock Supplier Portal for edge-case resolution and data-syncing.

A context-aware AI Chatbox for real-time document and ingredient assistance.

Stateful persistence using localStorage to simulate a database for the MVP demo.

Bilingual UI toggle (English and Bahasa Melayu).

7. Features Not Included (Scope Control)

- Implementation of a production backend database (e.g., PostgreSQL, Supabase).
- Cloud storage for actual document blob archiving (simulated client-side).
- Live direct API submission to the actual JAKIM MYEHALAL government portal.
- Automated email/WhatsApp dispatching to suppliers (simulated via UI state updates).

8. Assumptions & Constraints

- **LLM Cost Constraint:** We estimate the token cost per average user session to be extremely low (<$0.01). To keep inference costs manageable, we do not continuously poll the AI. API calls are strictly action-driven (e.g., only triggered when the user explicitly clicks "Verify" or sends a chat message).

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- **Technical Constraints:** Document processing is limited to text-extractable files (TXT, simple PDF, CSV) processed locally via the browser. Highly complex, image-heavy PDFs (requiring advanced OCR) are not supported in this prototype.
- **Performance Constraints:** Processing a massive ingredient list (e.g., 500+ items) in a single prompt may cause context degradation or high latency. The system assumes a standard SME recipe profile of 10-50 ingredients.
- **User Input:** The model acts as an intelligent assistant, not a regulatory authority. Submissions cannot be exported if AI confidence is low; it strictly requires human approval and supplier validation for edge cases.

9. Risks & Questions Throughout Development

- **AI Hallucination:** ***How do we ensure the AI doesn't accidentally label a Haram ingredient as Halal? * Mitigation:*** **Implementation of the "GLM Reasoning Audit Trail" (a UI panel showing the AI's exact logic) and strict confidence thresholds that force manual human review for anything under 60% certainty.**
- **State Stability & Demo Risk:** ***Does the user lose their entire application progress if the browser crashes or refreshes midway through the workflow? Mitigation: Robust implementation of React useEffect and localStorage to continuously cache workflow states, ensuring zero data loss during judging presentations.***

- **Regulatory Schema Changes:** ***What happens if JAKIM updates their required document list or data formatting rules? Mitigation: AI validation rules are not hardcoded into rigid if/else statements. They are defined via flexible Zod schemas and arrays, meaning the engine can adapt to new regulations by simply updating a single array variable.***

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